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RESEARCH

Recognizing uric acid type of urinary stones by deep learning

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Abstract

The purpose of this study was to investigate the effectiveness of deep learning (DL) methods in recognizing uric acid types from images of urinary stones. We curated a single-center cohort of 208 anonymized CT images from patients with urinary calculi. Images were rigorously screened (single calculus, diameter greater than or equal to 3 mm, high quality without artifacts), uniformly preprocessed to 512×512, and selected at the patient level (one image per patient) to prevent data leakage. Stones were labeled by Hounsfield Units from clinical reports into three classes—uric acid (0-600 HU), mixed (600–1000 HU), and non-uric acid (greater than 1000 HU)—yielding 50/72/86 images, respectively. We trained and validated these 208 urinary stone images using 7 deep learning (DL) networks and compared classification efficiency. Among seven deep learning models evaluated on the same curated dataset (n=208), ResNet18 yielded the best performance (accuracy=0.9808, precision=0.9828, recall=0.9761, F1=0.9790) and was selected as the final model for urinary stone composition prediction. At the argmax point, class-wise recalls were 94.00% (UA), 98.84% (NUA), and 100% (MIX); corresponding AUCs were 0.984, 0.987, and 0.977 (macro AUC=0.985), and APs were 0.974, 0.983, and 0.952 (macro AP=0.970). Sensitivity/specificity reached 0.94/1.00 (UA), 0.9884/0.9918 (NUA), and 1.00/0.9779 (MIX), indicating no missed MIX cases and no false positives for UA. Decision curve analysis (DCA) and clinical impact curves (CIC) showed positive net benefit across clinically plausible thresholds (notably around $P_t = 0.4$), suggesting efficient capture of true cases with a manageable screening burden. This study suggests that a ResNet18-based deep learning model may support highly accurate, preoperative, noninvasive identification of uric acid stones and has potential clinical utility. The approach has the potential to streamline the diagnostic-therapeutic pathway for urolithiasis and to reduce unnecessary invasive procedures. Future multicenter, prospective investigations will evaluate its real-world generalizability and cost-effectiveness, and explore multimodal data fusion to further improve recognition of complex stones. Our code and datasets are available at https://github.com/wbin2002/UrinaryStone.git.

Keywords Urinary stones · Uric acid · Deep learning

Introduction

Deep learning (DL), a core branch of artificial intelligence (AI), has achieved substantial progress in medical image analysis by learning hierarchical representations from imaging data and automatically extracting discriminative features for detection and classification tasks [1–3]. A typical DL workflow comprises training—where preprocessed data are used to optimize network parameters and learn task-specific features—and inference—where the learned weights are applied to unseen data for prediction. In recent years, DL-based pattern recognition has been widely adopted for computer-aided diagnosis, supporting clinicians in disease screening, lesion characterization, and clinical decision-making; in certain applications, DL systems have

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approached or surpassed radiologist-level performance in detection, illustrating the promise of human–computer collaboration to improve diagnostic accuracy and efficiency.

Urolithiasis is among the most common disorders of the urinary system and arises from a combination of environmental and genetic factors [4]. Its prevalence has continued to rise worldwide, with estimated rates of 1–5% in Asia, 5–9% in Europe, and 7–13% in North America [5]. According to the International Morphological Classification of Urinary Calculi [6], major compositional types include calcium oxalate monohydrate (COM), carbonate apatite (CA), magnesium ammonium phosphate hexahydrate (MAPH), and uric acid (UA). Moreover, nearly half of urinary stones exhibit a mixed morphology containing two or more constituents, as reported by Corrales et al. [7].

A substantial body of work has investigated Hounsfield unit–based metrics for stone assessment [8–15]. Early studies, such as Shahnani et al. [8] in a cohort of 180 patients with pure stones, examined the relationships between composition and both HU and Hounsfield density (HD), demonstrating significant associations. Gücük et al. [9] reviewed the clinical utility of HU and HD for evaluation and treatment, concluding that HU is a validated predictor of composition and extracorporeal shock wave lithotripsy (ESWL) outcomes, and may inform decision-making for percutaneous nephrolithotomy (PCNL). More recently, Plata et al. [10] analyzed 30 mixed stones using HU and HD to infer composition and found that anhydrous uric acid (367–556 HU) differs significantly from other types. Subsequent studies have leveraged HU and HD to differentiate stone types [11], extended HU-based analysis to microbiological prediction [12], and explored HU-related measures for prognosis and treatment response [13–15].

With the advances in minimally invasive and optical technologies, as well as the introduction of endoscopic devices, the treatment of urinary stone disease has become less painful for patients, thereby benefiting a larger population of individuals affected by this condition [16–22]. Current treatment strategies for urinary stones include extracorporeal shock wave lithotripsy [16] (ESWL), percutaneous nephrolithotomy [17] (PCNL), ureteroscopy [17], and conservative or pharmacological management depending on stone type and burden. ESWL remains a cornerstone technique, particularly effective for renal and ureteral stones; however, its efficacy is limited for certain compositions such as uric acid stones, and it may cause renal injury, functional impairment, and an increased risk of recurrence [18]. PCNL is an effective intervention for large, multiple, or staghorn renal calculi and is widely adopted in clinical practice [19]. Nevertheless, as an invasive procedure, PCNL carries the risk of irreversible tissue damage and postoperative complications. Ureteroscopy provides another therapeutic option, offering effective treatment of intraluminal stones with reduced trauma and improved outcomes in complex cases [20]. However, due to its technical characteristics, this method is associated with risks such as ureteral injury and urinary tract infection. Consequently, accurate preoperative determination of stone composition is of great clinical significance, as it directly guides the selection of appropriate treatment strategies and helps avoid unnecessary or ineffective interventions.

Urolithiasis frequently exhibits mixed chemical composition, with purely single-component stones representing only a minority of cases [23]. Black et al. [24] demonstrated the feasibility of deep learning for stone composition prediction using a ResNet-101 convolutional neural network on 63 stones (e.g., calcium oxalate monohydrate, uric acid), but their analysis was limited to pure stones and therefore does not reflect the mixed presentations common in clinical practice. Building on this line of work and related evidence that deep learning can assist composition analysis from CT and routine laboratory data [25], the present study undertakes a systematic comparison of seven convolutional architectures for multi-class prediction of urinary stone composition from CT images, explicitly including mixed stones. Our objective is to develop and select a noninvasive, preoperative model that achieves high discriminative performance and demonstrates clinical utility—particularly for identifying uric acid stones, which are amenable to medical dissolution—thereby supporting urologists' decision-making and reducing unnecessary invasive procedures.

Materials and methods

Datasets

The experimental data in this study were provided by the First Affiliated Hospital of Ningbo University, comprising computed tomography (CT) images from 208 patients diagnosed with urinary calculi between January 2025 and August 2025. To ensure both sample quality and cross-experimental comparability, all candidate images underwent a rigorous screening process, and only those meeting the following criteria were retained: (i) stone diameter no less than 3 mm, (ii) single calculus cases (excluding patients with multiple stones), and (iii) high-quality images without apparent artifacts. In addition, all CT images were anonymized prior to use, with any information that could potentially identify patient identity removed. The images used for the experiments were uniformly set to a resolution of 512×512 pixels and contained no privacy-related identifiers. To prevent data leakage, we adopted a patient-level selection strategy, whereby only one CT image per patient was included in the dataset, thus avoiding the presence of multiple images from the same individual in both training and test sets.

During data preprocessing, raw CT images in DICOM format were exported from Siemens devices and converted into JPG format by professional radiologists. The images were then cropped and resized to 512 × 512 pixels to match the input requirements of the subsequent deep learning models. Furthermore, based on the Hounsfield Units (HU) reported in the corresponding clinical CT examination records, the stone images were categorized into three classes: 0–600 HU as uric acid stones, 600–1000 HU as mixed stones, and more than 1000 HU as non-uric acid stones. Ultimately, the curated dataset consisted of 208 CT images distributed across three categories: 50 uric acid stones, 72 mixed stones, and 86 non-uric acid stones (Table 1).

Method

All experiments were conducted on a Matpool cloud server equipped with an Intel® Xeon® Platinum 8352 V CPU and an NVIDIA GeForce RTX 4090 GPU. A Conda environment was created on Ubuntu 20.04 with Python 3.10, PyTorch 2.4.0, and CUDA 11.8. Moreover, we used a batch size of 8, trained for 50 epochs, set the initial learning rate to 1e−3, and employed 4 data-loading workers. No dedicated denoising algorithm was applied in this study, as all CT images were acquired under standard clinical protocols with routine noise control. To reduce variability across scans, all images were resampled to a uniform size. These preprocessing steps ensured consistency of the input data while retaining clinically relevant structural details. Moreover, to mitigate class imbalance and reduce overfitting, we applied class-preserving data augmentation only during training. Each input CT image was subjected to a stochastic sequence of geometric and photometric transforms sampled on the fly at every epoch: (i) random resized crop to 224 × 224 pixels; (ii) horizontal flip with probability $p = 0.5$; (iii) in-plane rotation uniformly sampled from [−15°,+15°]; and (iv) color jitter with brightness, contrast, and saturation each perturbed by up to ± 20%. These augmentations are designed to emulate real-world variations (viewpoint, positional shift, acquisition angle, and imaging conditions), thereby enriching the effective training distribution and encouraging the model to learn more generalizable representations. No augmentation was applied to the test sets, which were only resized to 224 × 224 to ensure consistent evaluation.

Table 1 Dataset of urinary stones

Class	HU Range	Images
Uric-acid	0–600 HU	50
Mixed	600–1000 HU	72
Non-uric-acid	> 1000 HU	86

Model development and evaluation followed a patient-level Leave-One-Out (LOO) protocol to prevent data leakage across splits. Specifically, folds were defined by unique patient identifiers; at each iteration, one patient's CT image served as the test set while all remaining patients' images were used for training. This design preserves clinical validity by ensuring that no subject contributes data to both training and test sets.Within each fold, the backbone was initialized with pretrained weights, and the final fully connected layer was replaced to produce three-class outputs. Training used cross-entropy loss optimized with Adam, together with a step learning-rate scheduler (multiplicative factor 0.1 every 10 epochs). Validation accuracy was monitored throughout training, and the checkpoint with the best validation performance was retained to mitigate overfitting.

Stone composition in this study was determined by preoperative CT-based Hounsfield Unit (HU) thresholds rather than postoperative chemical analysis, because infrared spectroscopy or X-ray diffraction (XRD) was not routinely performed for all retrieved stones at our institution. HU-based classification remains a widely used noninvasive clinical surrogate for stone type in many centers where stone retrieval or chemical analysis is not feasible. Nevertheless, HU measurements can be influenced by scanner type, acquisition parameters (kVp, mAs), reconstruction kernel or algorithm, and patient body habitus, which may introduce systematic bias or misclassification - particularly in mixed stones or stones with HU values near classification thresholds. Therefore, our results should be interpreted in light of this limitation, and future validation employing chemical composition as the gold standard is warranted.

Evaluation protocol

We evaluate classification performance using accuracy, precision, recall, and the F1 score. Accuracy measures the overall proportion of correct predictions; in the multi-class setting, its numerator is the sum of the diagonal entries of the confusion matrix and the denominator is the total number of samples. Precision denotes, among all samples predicted as positive, the fraction that are truly positive; recall denotes, among all truly positive samples, the fraction correctly identified. The F1 score—the harmonic mean of precision and recall—provides a robust summary under class imbalance.

For the three-class setting, let the confusion matrix be $C\epsilon\ N^{3X3}$, where C_{ij} is the number of samples of true class i predicted as class j, and $N = \sum_{i,j} C_{ij}$ is the total sample size. Under a one-vs-rest scheme for class $K\epsilon UA,\ NUA,\ MIX$, $TP_k = C_{kk}$, $FP_k = \sum\limits_{i \neq k} C_{ik}$, $FN_k = \sum\limits_{j \neq k} C_{kj}$, $TN_k = N - TP_k - FP_k - FN_k$.

The metrics are defined as.

$$Accuracy = \frac{\sum_k C_{kk}}{N}$$

$$\Pr ecision_k = \frac{TP_k}{TP_k + FP_k}$$

$$\operatorname{Re} call_k = \frac{TP_k}{TP_k + FN_k}$$

$$F1\_score_k = \frac{2 \Pr ecision_k \cdot \operatorname{Re} call_k}{\Pr ecision_k + \operatorname{Re} call_k} = \frac{2TP_k}{2TP_k + FP_k + FN_k}$$

When a single summary value is desired, we report macro-averaged metrics:

$$M_{macro} = \frac{1}{3} \sum_k M_k \cdots M \in \{\Pr ecision, \operatorname{Re} call, F1\_score\}$$

In addition, to quantify the model's detection and exclusion performance at the operating point defined by the argmax decision rule, we report sensitivity and specificity.

$$Sensitivity_k = \frac{TP_k}{TP_k + FN_k}$$

$$Specificity_k = \frac{TN_k}{TN_k + FP_k}$$

Moreover, we quantified the clinical utility of the classifier using decision curve analysis (DCA). For a threshold probability $p_t \in (0, 1)$, the net benefit (NB) is defined as

$$NB(p_t) = \frac{TP}{N} - \frac{FP}{N} \cdot \frac{p_t}{1 - p_t}$$

where N is the sample size, and TP and FP denote the numbers of true and false positives at the operating threshold. The factor $\frac{p_t}{1-p_t}$ encodes the relative harm of a false positive versus the benefit of a true positive.

We further plotted clinical impact curves (CIC) to visualize patient-level consequences across decision thresholds. For each threshold probability p_t , the CIC reports (i) the number of individuals classified as high risk (predicted positive) and (ii) the expected number of true positives among them. Formally,

$$HighRisk(p_t) = \hat{P}(\hat{y} = 1|p_t) \cdot N$$

$$TP(p_t) = \hat{P}(\hat{y} = 1, y = 1|p_t) \cdot N$$

where N is the sample size. In the multi-class setting, we adopted a one-vs-rest scheme per class. For comparability across studies, we normalized by the test-set size and reported curves over the same P_t grid as DCA.

Results

We conducted a comparative evaluation to assess the efficiency and accuracy of seven deep learning architectures-MobileNetV3Small, ResNet18, ResNet50, VGG11, VGG16, ShuffleNetV2, and EfficientNetV2B0. Each model was trained and validated on the same curated dataset selected under the predefined screening criteria, using an identical training protocol. To quantify the uncertainty of our performance estimates, we performed 1,000 bootstrap resamples under the LOOCV framework and computed 95% confidence intervals (CIs) for all primary metrics. The resulting CIs are presented in Table 2. Among these candidates, ResNet18 achieved the best overall performance, with an accuracy of 0.981 (95% CI: 0.962–0.995), precision of 0.983 (95% CI: 0.964–0.996), recall of 0.976 (95% CI: 0.953–0.996), and F1 score of 0.979 (95% CI: 0.957–0.996). Accordingly, ResNet18 was adopted as our final model for predicting urinary stone composition.

We evaluated the ResNet18 model using the confusion matrix, receiver operating characteristic (ROC) curves, and precision-recall (PR) curves (Figs. 1.(a)-(c)). Figure 1.(a) presents the confusion matrix for three categories—uric acid (UA), non-uric acid (NUA), and mixed (MIX). At the argmax operating point, the class-wise recalls of ResNet18 are 94% (UA), 99% (NUA), and 100% (MIX). Specifically, among 50 UA images, 1 was misclassified as NUA and 2 as MIX; among 86 NUA images, 1 was misclassified as MIX; all 72 MIX images were correctly classified. Figure 1(b) presents the one-vs-rest ROC curves, indicating strong class separability with AUCs of 0.984 for UA (95% CI: 0.963–1.000.963.000), 0.987 for NUA (95% CI: 0.976–0.996), and 0.977 for MIX (95% CI: 0.958–0.991). The macro-averaged AUC is 0.985 (95% CI: 0.973–0.993), clearly exceeding the random baseline of 0.5. Figure 1.(c) shows the one-vs-rest PR curves. The model maintains high average precision (AP) across classes—0.974 (UA), 0.983 (NUA), and 0.952

Table 2 Performance comparison of deep learning models on urinary stone composition classification (Macro-Averaged Accuracy, Precision, Recall, F1-Score, and AUC With 95% Confidence Intervals)

Model	Accuracy 95% CI	Precision 95% CI	Recall 95% CI	F1 score 95% CI	AUC 95% CI
MobileNetv3_small	0.938 (0.899,0.961)	0.937 (0.915,0.970)	0.927 (0.897,0.958)	0.931 (0.905,0.963)	0.962 (0.946,0.977)
ResNet18	**0.981 (0.962,0.995)**	**0.983 (0.964,0.996)**	**0.976 (0.953,0.996)**	**0.979 (0.957,0.996)**	**0.985 (0.973,0.993)**
ResNet50	0.947 (0.918,0.976)	0.951 (0.921,0.977)	0.939 (0.904,0.969)	0.944 (0.911,0.973)	0.979 (0.967,0.988)
Vgg11	0.846 (0.793,0.889)	0.886 (0.848,0.917)	0.791 (0.742,0.835)	0.791 (0.724,0.847)	0.885 (0.845,0.918)
Vgg16	0.870 (0.827,0.913)	0.893 (0.852,0.931)	0.826 (0.779,0.874)	0.835 (0.775,0.885)	0.883 (0.841,0.921)
ShuffleNetv2	0.894 (0.851,0.938)	0.885 (0.837,0.932)	0.877 (0.828,0.923)	0.881 (0.833,0.924)	0.907 (0.887,0.927)
EfficientNetv2_b0	0.952 (0.918,0.976)	0.955 (0.926,0.980)	0.937 (0.896,0.970)	0.944 (0.907,0.973)	0.955 (0.937,0.970)

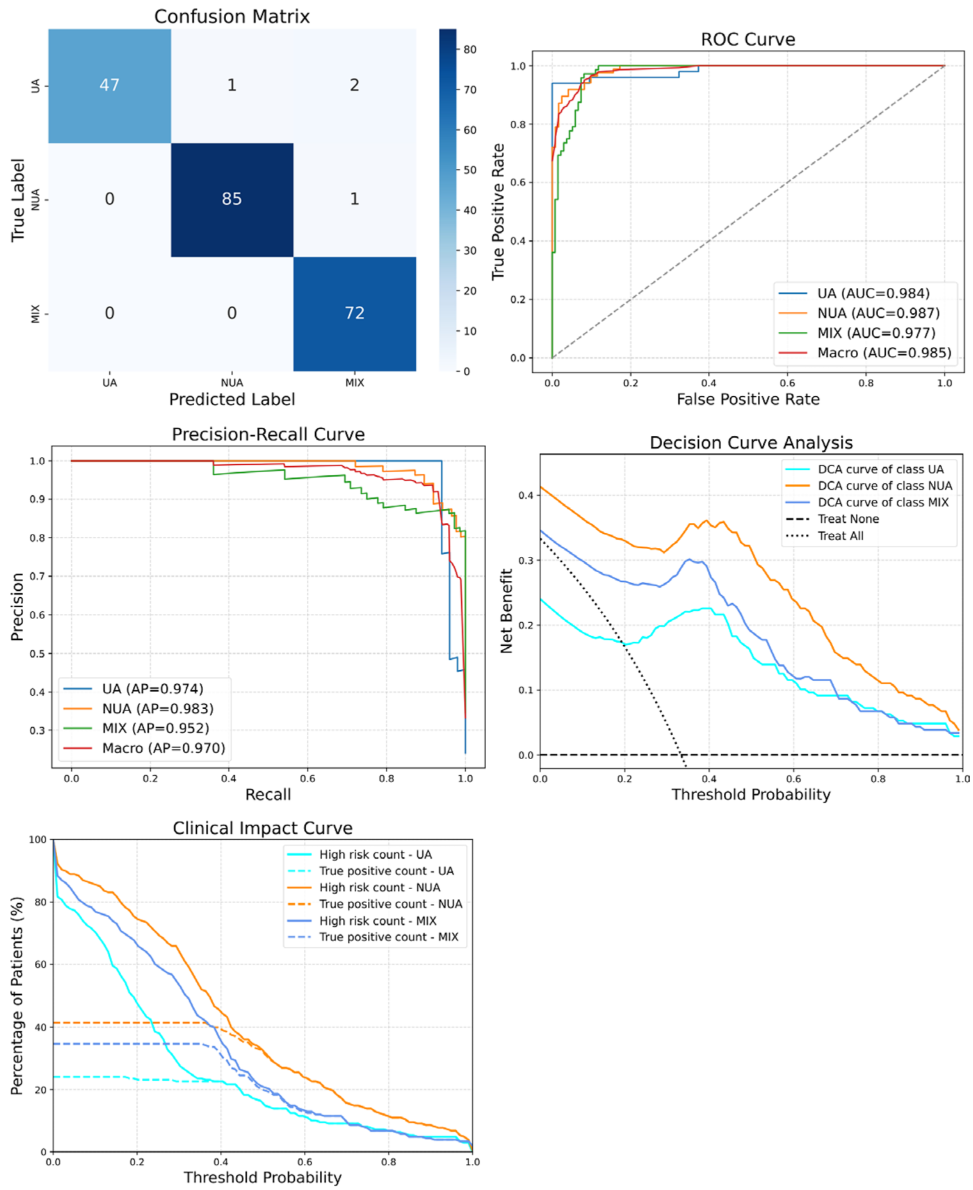

Fig. 1 (**a**)Confusion matrix of the ResNet18 model for the three-class task (UA/NUA/MIX). Diagonal entries indicate correct predictions; the overall accuracy is 98.08% (204/208). Experimental results and analysis of Resnet18. (**b**) One-vs-rest ROC curves for uric acid, (UA), non-uric acid, (NUA), and mixed, (MIX) stones obtained with the ResNet18 model. The diagonal denotes random performance, (AUC = 0.5)., (**c**) One-vs-rest Precision–Recall, (PR) curves for uric acid, (UA), non-uric acid, (NUA), and mixed, (MIX) stones obtained with the ResNet18 model., (**d**) One-vs-rest decision curve analysis, (DCA) for the ResNet18 model. The curves depict the net benefit as a function of the threshold probability for uric acid, (UA), non-uric acid, (NUA), and mixed, (MIX) stones. The black dashed and dotted lines denote the "Treat None" and "Treat All" strategies, respectively., (**e**) Clinical impact curves, (CIC) for the ResNet18 model under a one-vs-rest scheme. The solid lines show the proportion classified as high risk, and the dashed lines indicate the expected proportion of true positives

Table 3 Class-wise sensitivity and specificity at the argmax operating point for the three-class task (UA/NUA/MIX). Macro-averaged sensitivity and specificity are 0.976 and 0.990, respectively

Class	Sensitivity 95% CI	Specificity 95% CI
UA	0.940 (0.840,0.980)	1.000 (1.000,1.000)
NUA	0.988 (0.945,1.000)	0.992 (0.966,1.000)
MIX	1.000 (1.000,1.000)	0.978 (0.945,0.995)

(MIX)—with a macro AP of 0.970, indicating sustained precision at clinically relevant high-recall levels. Notably, as recall approaches 1.0, the MIX curve exhibits an earlier precision drop than UA/NUA, suggesting greater overlap between MIX and the other categories.

At the operating point corresponding to the argmax decision rule, the class-wise sensitivities and specificities of the ResNet18 model were 0.940 (95% CI: 0.840–0.980) and 1.000 (95% CI: 1.000–1.000) for uric acid (UA), 0.988 (95% CI: 0.945–1.000.945.000) and 0.992 (95% CI: 0.996–1.000.996.000) for non-uric-acid (NUA), and 1.000 (95% CI: 1.000–1.000) and 0.978 (95% CI: 0.945–0.995) for mixed stones (MIX), respectively (Table 3). The MIX class achieved a sensitivity of 1.000, accompanied by a modest reduction in specificity, whereas the UA class reached a specificity of 1.000. The NUA class showed a more balanced profile, with both sensitivity and specificity exceeding 0.988. The macro-averaged sensitivity and specificity were 0.976 and 0.990, respectively, which is consistent with the overall performance observed in the confusion matrix and the ROC/PR analyses.

We further assessed the clinical utility of ResNet18 using decision curve analysis (DCA) and clinical impact curves (CIC). Figure 1.(d) presents the one-vs-rest DCA results. Across threshold probabilities $p_t \in [0.2, 0.6]$, all three DCA curves lie above the Treat None and Treat All strategies, indicating positive net benefit when the model is used for decision making. The NUA class achieves the widest range and largest magnitude of net benefit, followed by MIX and UA. The net benefit peaks for all three classes at approximately $p_t \approx 0.4$. Figure 1.(e) shows the CIC results, where at $p_t \approx 0.4$ the proportion of patients flagged as high risk closely matches the class-specific proportions of true positives, suggesting that, at this threshold, the model captures the majority of true cases with a manageable screening burden. To facilitate clinical interpretation, we relate the threshold probability values to practical treatment decisions. For UA stones, the model demonstrated a positive net benefit within a threshold probability range of approximately 0.3–0.5. This suggests that when the predicted probability of UA reaches moderate levels (around 30–50%), acting upon the model output—such as considering chemolysis for suspected uric-acid-dominant stones—may provide greater overall benefit than treating all patients or treating none. For NUA stones, the net-benefit advantage extended across a wider threshold range (0.3–0.6), indicating that model-guided selection of ESWL or other non-chemolytic strategies could be advantageous within this interval. For mixed-composition stones, a similar moderate threshold range (approximately 0.3–0.6) yielded positive net benefit, reflecting the potential value of management strategies that address hard-stone components while avoiding unnecessary chemolysis. Collectively, these threshold ranges correspond to clinically plausible decision preferences and illustrate how the model may assist in stone-type-specific treatment selection. The net-benefit and cumulative incidence curves thus highlight decision thresholds at which the model could meaningfully influence clinical management and potentially reduce inappropriate interventions.

Discussion

Deep learning, a leading branch of machine learning, leverages deep neural architectures to automatically extract features and build hierarchical representations from large-scale data. Enabled by GPU parallelism and distributed training frameworks, it has achieved notable progress in medical image analysis in recent years, opening new avenues for clinical applications of artificial intelligence [26, 27]. Nevertheless, successful deployment in routine care, particularly for critical diagnostic tasks such as stone composition identification, still faces key challenges, including the acquisition of high-quality labeled data and model interpretability.

This study employed the ResNet18 architecture to extract imaging features beyond conventional Hounsfield units (HU), thereby addressing the high misclassification rates in traditional CT imaging caused by overlapping stone compositions. With its relatively excellent discriminative capability (macro AUC = 0.985), the proposed approach enables non-invasive preoperative identification of uric acid stones, marking an important step toward 'preoperative precision prediction' of stone composition. Clinically, accurate determination of urinary stone composition is critical in urology, as different stone types necessitate distinct treatment strategies. While conventional laboratory techniques such as infrared spectroscopy and chemical analysis can provide reliable compositional information, they are time-consuming, labor-intensive, and costly, and may not be readily accessible in all clinical settings. The findings of this study indicate that deep learning (DL)-based classification models, particularly ResNet18, may serve as a rapid and cost-effective adjunct to these traditional methods. Such models can assist clinicians in selecting appropriate interventions at an earlier stage, thereby preventing uric acid stone patients from undergoing ineffective extracorporeal shock wave lithotripsy (ESWL) and supporting timely initiation of pharmacological dissolution therapy. Specifically, for patients with uric acid stones, the model reduced misclassification risk by approximately two-thirds, thereby improving therapeutic efficiency, minimizing procedure-related risks and costs, and providing evidence (level Ib) to support standardized non-surgical management. Furthermore, automated classification decreases reliance on manual assessment, reducing inter-observer variability and alleviating the workload of laboratory staff.

The study also reveals limitations. Performance for mixed stones (MIX) is lower than for pure types (AUC 0.977), primarily due to the relatively small number of MIX samples (n = 72), which constrains the model's ability to learn the class's heterogeneous imaging patterns. Such class imbalance is common in related work [6, 28]. Liu et al. [28] proposed a self-distillation framework for noninvasive prediction of urinary stone composition; however, pronounced imbalance and small sample sizes for certain classes limited feature learning and affected outcomes. Estrade et al. [6] evaluated a deep recurrent network on a dataset with approximately 80% pure and approximately 20% mixed stones and reported higher accuracy for pure stones, consistent with our observations. In a previous study [29], clinician visual assessment of CT images was evaluated using chemical analysis as the reference standard, and reported an overall accuracy of 39.1%. Whereas in our study, stone type was determined based on HU thresholds. Because these two reference standards differ fundamentally, a direct numerical comparison of diagnostic accuracy is not appropriate. Instead, our intention was to highlight the widely recognized difficulty of visually distinguishing stone composition on CT, particularly for mixed stones, as reported in recent studies showing substantial inter-reader variability and limited accuracy using visual estimation alone [30, 31]. Our intention is to emphasize that imaging-based stone characterization remains challenging in routine practice, and that automated HU-driven deep learning approaches may complement clinical judgement by providing standardized, reproducible decision support.

Although deeper networks such as ResNet50 and EfficientNetV2_b0 usually achieve superior performance in large-scale visual recognition tasks, our results indicate that ResNet18 outperformed the other evaluated architectures in urinary stone composition classification. A possible explanation is that ResNet18 strikes a balance between network depth and parameter efficiency. Its relatively smaller number of parameters reduces the risk of overfitting when trained on a limited dataset, while the residual connections effectively facilitate feature propagation and gradient flow. Moreover, the moderate depth of ResNet18 enables efficient extraction of discriminative features without introducing excessive computational complexity, making it particularly suitable for the present clinical imaging dataset.

All CT images analyzed in this study were acquired at a single center using three scanners under a uniform acquisition protocol (180 kV, 200 mAs). Consequently, imaging-parameter diversity and patient demographic variability were limited, which may restrict the generalizability of our findings. Previous studies have demonstrated that CT attenuation values (HU) and derived radiomic features are highly sensitive to acquisition and reconstruction settings. Notably, inter- and intra-scanner variations, differing kVp/mAs settings, reconstruction kernels, and post-processing algorithms can produce systematic shifts in HU and texture features, thereby impairing reproducibility across institutions [32–35]. Indeed, radiomics reproducibility studies have reported that only a minority of features remain stable when such parameter variations occur. In our context, since our stone classification model relies on HU-based labels and network learning from HU distributions, there is a risk that the model may overfit to scanner-/protocol-specific intensity distributions rather than stone-intrinsic properties. Therefore, the model's performance might decline when applied to external data acquired with different scanners, protocols, or reconstruction settings. To address this, future work will involve external validation on multi-center datasets with broader imaging and patient heterogeneity, and the application of harmonization or domain-adaptation techniques (e.g., ComBat/CovBat-based radiomics harmonization) to mitigate scanner- and protocol-related variability and improve cross-center robustness.

This study did not include postoperative chemical analysis (e.g., infrared spectroscopy or X-ray diffraction) because such testing was not routinely performed for all patients at our institution. As a result, the HU-based stone type labels used in this study cannot fully replace the gold-standard biochemical determination of stone composition. Prior work has emphasized that chemical analysis remains essential for accurate stone characterization and is the definitive reference standard for validating imaging-based methods [36, 37]. Future multi-center studies incorporating routine postoperative chemical analysis are therefore required to rigorously evaluate the model's diagnostic performance.

This study is constrained by a single-center, retrospective cohort and a limited architectural search restricted to seven CNN backbones; although ResNet18 performed best, broader families (e.g., 3D CNNs, transformers, multimodal fusion) were not evaluated. No dedicated denoising algorithm was applied during preprocessing. Although all CT scans were acquired under standard clinical protocols with routine noise control, residual noise [38] and artifacts may still be present and could potentially affect the robustness and generalizability of the deep learning model. Finally, residual class imbalance [39] in this retrospective cohort, together with variability in stone size and compositional subtypes, may challenge generalizability.

Future studies should include prospective multi-center cohorts with greater diversity in scanners, acquisition parameters, and patient demographics, and incorporate postoperative chemical analysis (e.g., FTIR or XRD) to validate the HU-based reference standard. Efforts to mitigate domain shift—such as image harmonization, normalization, advanced augmentation, and reconstruction-aware or deep learning-based denoising—may help improve robustness across heterogeneous imaging conditions and address class imbalance. Finally, external validation on independent multi-center datasets, along with repeated experiments and uncertainty quantification, will be essential to assess reproducibility and clarify the model's potential clinical utility, particularly for challenging mixed-composition stones.

Conclusions

This study suggests that a ResNet18-based deep learning model may support highly accurate, preoperative, noninvasive identification of uric acid stones and has potential clinical utility. The proposed approach could help streamline the diagnostic-therapeutic pathway for urolithiasis and reduce unnecessary invasive procedures. Future multicenter, prospective investigations are warranted to assess its real-world generalizability and cost-effectiveness, as well as to explore multimodal data fusion to further improve the characterization of complex stones.

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Author contributions B.W.:Data Curation, Formal Analysis, Methodology, Software, Validation, Visualisation; Y.Z.:Data Curation, Formal Analysis, Methodology, Resources, Validation; J.C.: Conceptualisation, Funding Acquisition, Investigation, Methodology, Project Administration, Supervision, Validation. B.W., Y.Z., and J.C. wrote the main manuscript text. All authors reviewed the manuscript.

Data availability Our code and datasets are available at https://github.com/wbin2002/UrinaryStone.git. More data can be obtained by contacting the corresponding author.

Declarations

Competing interests The authors declare no competing interests.

Ethics approval This retrospective study was approved by the Ethics Committee of the First Affiliated Hospital of Ningbo University (approval No.: Ethics Review 2025–Research–164RS). Because the analysis used only pre-existing, de-identified clinical records and posed minimal risk, the IRB granted a waiver of informed consent. All procedures complied with the Declaration of Helsinki and applicable regulations.

References

1. Zhu S, Cai J, Xiong R et al (2025) Singular pooling: a spectral pooling paradigm for second-trimester prenatal level II ultrasound standard fetal plane identification. IEEE Trans Circuits Syst Video Technol 35(12): 12508-12523. https://doi.org/10.1109/TCSVT.2025.3588395
2. Cai J, Li Y, Liu B et al (2024) Developing deep LSTMs with later temporal attention for predicting COVID-19 severity, clinical outcome, and antibody level by screening serological indicators over time. IEEE J Biomed Health Inform 28(7):4204–4215. https://doi.org/10.1109/JBHI.2024.3384333
3. Zhu S, Cai J, Xiong R et al (2025) Contrast and gain-aware attention: a plug-and-play feature fusion attention module for torso region fetal plane identification. Ultrasound Med Biol 51(12):2258–2266. https://doi.org/10.1016/j.ultrasmedbio.2025.08.014
4. Amir A, Matlaga BR, Ziemba JB, Sheikh S (2018) Kidney stone composition in the Kingdom of Saudi Arabia. Clin Nephrol 89(5):345–348. https://doi.org/10.5414/CN109313
5. Song BI, Lee J, Jung W, Kim BS (2023) Pure uric acid stone prediction model using the variant coefficient of stone density measured by thresholding 3D segmentation-based methods: a multicenter study. Comput Methods Programs Biomed 240:107691. https://doi.org/10.1016/j.cmpb.2023.107691
6. Estrade V, Daudon M, Richard E, Bernhard JC, Bladou F, Robert G, de Denis Senneville B (2022) Towards automatic recognition of pure and mixed stones using intra-operative endoscopic digital images. BJU Int 129(2):234–242. https://doi.org/10.1111/bju.15515
7. Corrales M, Doizi S, Barghouthy Y, Traxer O, Daudon M (2021) Classification of stones according to Michel Daudon: a narrative review. Eur Urol Focus 7(1):13–21. https://doi.org/10.1016/j.euf.2020.11.004
8. Shahnani PS, Karami M, Astane B, Janghorbani M (2014) The comparative survey of Hounsfield units of stone composition in urolithiasis patients. J Res Med Sciences: Official J Isfahan Univ Med Sci 19(7):650
9. Gücük A, Üyetürk U (2014) Usefulness of Hounsfield unit and density in the assessment and treatment of urinary stones. World J Nephrol 3(4):282
10. Rodríguez-Plata IT, Medina-Escobedo M, Basulto-Martinez M, Avila-Nava A, Gutierrez-Solis AL, Mendez-Dominguez N, Lugo R (2021) Implementation of a technique based on Hounsfield units and Hounsfield density to determine kidney stone composition. Tomography 7(4):606–613
11. Sánchez-Pozos K, Ramírez-Jurado AA, Medina-Escobedo M, Garrido-Dzib ÁG, González-Rocha LA, Gutiérrez-Solis AL, Avila-Nava A, Lugo R (2024) Computed tomographic characterization of urinary stones in patients with urolithiasis from Southeast Mexico. Heliyon 10(1).e23547. https://doi.org/10.1016/j.heliyon.2023.e23547
12. Kalyenci B, Çift A, Aydemir AB, Sulhan H, Benlioğlu C, Yücel MÖ (2025) Investigation of the predictive value of Hounsfield units in predicting stone culture results in urinary stone disease. Sci Rep 15(1):16249
13. Garg M, Johnson H, Lee SM, Rai BP, Somani B, Philip J (2023) Role of Hounsfield unit in predicting outcomes of shock wave

lithotripsy for renal calculi: outcomes of a systematic review. Curr Urol Rep 24(4):173–185
14. Alharbi AS, Gameraddin M, Gareeballah A, Shrwani ZJ, Sindi MAA, Alsaedi HI, Alzain AF (2024) Assessment of Hounsfield units and factors associated with fragmentation of renal stones by extracorporeal shock wave lithotripsy: a computerized tomography study. Tomography 10(1):90–100
15. Moon HW, Taeyb M, Park YH, Bae WJ, Ha US, Hong SH, Cho HJ (2022) The impact of Hounsfield unit-related variables on percutaneous nephrolithotomy outcomes. Sci Rep 12(1):18451
16. Oliveira B, Teixeira B, Magalhães M, Vinagre N, Fraga A, Cavadas V (2024) Extracorporeal shock wave lithotripsy: retrospective study on possible predictors of treatment success and revisiting the role of non-contrast-enhanced computer tomography in kidney and ureteral stone disease. Urolithiasis 52(1):65
17. Ellison JS, Chu DI, Nelson CP, DeFoor WR, Ziemba J, Huang J, Meenakshi-Sundaram B (2025) Percutaneous nephrolithotomy vs ureteroscopy for kidney stones in children. JAMA Netw Open 8(6):e2516749–e2516749
18. Yi JH, Li ZS, Hu LH (2023) Adverse events of pancreatic extracorporeal shock wave lithotripsy: a literature review. BMC Gastroenterol 23(1):360. https://doi.org/10.1186/s12876-023-02992-0
19. Hou J, Lv Z, Wang Y, Wang X, Wang Y, Wang K (2023) Knowledge-map analysis of percutaneous nephrolithotomy (PNL) for urolithiasis. Urolithiasis 51(1):34. https://doi.org/10.1007/s00240-023-01406-w
20. He M, Dong Y, Cai W, Cai J, Xie Y, Yu M, Li C, Wen L (2024) Recent advances in the treatment of renal stones using flexible ureteroscopys. Int J Surg (London England) 110(7):4320–4328. https://doi.org/10.1097/JS9.0000000000001345
21. Fu Z et al (2021) The future of endoscopic navigation: a review of advanced endoscopic vision technology. IEEE Access 9:41144–41167. https://doi.org/10.1109/ACCESS.2021.3065104
22. He Z, Wang P, Ye X (2021) Novel endoscopic optical diagnostic technologies in medical trial research: recent advancements and future prospects. BioMed Eng Online 20:5. https://doi.org/10.1186/s12938-020-00845-5
23. Kim US, Kwon HS, Yang W, Lee W, Choi C, Kim JK, Lee SH, Rim D, Han JH (2022) Prediction of the composition of urinary stones using deep learning. Investigative and clinical urology 63(4):441–447. https://doi.org/10.4111/icu.20220062
24. Black KM, Law H, Aldoukhi A, Deng J, Ghani KR (2020) Deep learning computer vision algorithm for detecting kidney stone composition. BJU Int 125(6):920–924. https://doi.org/10.1111/bju.15035
25. Jin L, Chen Z, Sun Y, Tian Z, Yi X, Huang Y (2024) Advancements in uric acid stone detection: integrating deep learning with CT imaging and clinical assessments in the upper urinary tract. Urol Int 108(3):234–241. https://doi.org/10.1159/000538133
26. Siddique M, Ashour MW Parallel Computing with GPU: An accelerator for Data-Centric High Performance Computing,2024 1st International Conference on Innovative Engineering Sciences and Technological Research (ICIESTR), Muscat, Oman, 2024, pp. 1–6. https://doi.org/10.1109/ICIESTR60916.2024.10798202
27. Rakhimov M, Javliev S, Nasimov R (2023) Parallel Approaches in Deep Learning: Use Parallel Computing. In ACM International Conference Proceeding Series(pp. 192–201). Association for Computing Machinery. https://doi.org/10.1145/3644713.3644738
28. Liu K, Zhang X, Yu H et al (2024) Efficient urinary stone type prediction: a novel approach based on self-distillation. Sci Rep 14:23718. https://doi.org/10.1038/s41598-024-73923-6
29. Sampogna G, Basic D, Geavlete P, Galán Llopis JA, Reis Santos J, Saltirov I, Sarica K, Stavridis S, Skolarikos A, Trinchieri A (2021) Endoscopic identification of urinary stone composition: A study of South Eastern Group for Urolithiasis Research (SEGUR 2). Identificación endoscópica de la composición de los cálculos urinarios: un estudio del Southeastern Group for Lithiasis Research (SEGUR 2). Actas Urol Esp 45(2):154–159. https://doi.org/10.1016/j.acuro.2020.01.015
30. Henderickx MM, Stoots SJ, De Bruin DM, Wijkstra H, Freund JE, Wiseman OJ, Henderickx MMEL, Stoots SJM, Ploumidis A, Skolarikos A, Somani BK, Şener TE, Emiliani E, Dragos LB, Villa L, Talso M, Daudon M, Traxer O, Kronenberg P, Doizi S, Tailly T, Tefik T, Hendriks N, Beerlage HP, Baard J, Kamphuis GM (2022) How reliable is endoscopic stone recognition? A comparison between visual stone identification and formal stone analysis. J Endourol 36(10):1362–1370
31. Rasali, N., Shrestha, A., Shrestha, P. M., Basnet, R. B., Adhikari, B., Shah, A. K.,… Thapa, B. B. (2025). Agreement Between Visual Assessment of Urinary Stones Morphology During Endo-Urological Procedure and Fourier Transform Infrared Spectroscopy-Based Composition Analysis: An Observational Study. Journal of the Nepal Medical Association, 63(290)
32. Jahanshahi A, Soleymani Y, Fazel Ghaziani M, Khezerloo D (2023) Radiomics reproducibility challenge in computed tomography imaging as a nuisance to clinical generalization: a mini-review. Egypt J Radiol Nucl Med 54(1):83
33. Soleymani Y, Valibeiglou Z, Ghaziani MF, Jahanshahi A, Khezerloo D (2024) Radiomics reproducibility in computed tomography through changes of ROI size, resolution, and Hounsfield unit: a phantom study. Radiography 30(6):1629–1636
34. Peng X, Yang S, Zhou L, Mei Y, Shi L, Zhang R, Liu L (2022) Repeatability and reproducibility of computed tomography radiomics for pulmonary nodules: a multicenter phantom study. Invest Radiol 57(4):242–253
35. Zhang H, Lu T, Wang L, Xing Y, Hu Y, Xu Z, Zhong J (2025) Robustness of radiomics within photon-counting detector CT: impact of acquisition and reconstruction factors. Eur Radiol. https://doi.org/10.1007/s00330-025-11374-x
36. Siener R, Herwig H, Rüdy J, Schaefer RM, Lossin P, Hesse A (2022) Urinary stone composition in germany: results from 45,783 stone analyses. World J Urol 40(7):1813–1820
37. Basiri A, Tahvildari A, Naji M, Ziaeefar P, Kashi AH (2025) Determination of the kidney stone composition using infrared spectroscopy in Iran at a National referral center during 2019–2023. Asian J Urol 12(1):72–78
38. Cai J, Qi Y, Liu S et al (2025) A residual joint antenna network for joint transmit–receive antenna subset selection in MIMO systems. IEEE Trans Antennas Propag 73(10):8366–8371. https://doi.org/10.1109/TAP.2025.3581721
39. Cai J, Chen T, Qi Y et al (2025) Fibrosis and inflammatory activity diagnosis of chronic hepatitis C based on extreme learning machine. Sci Rep15,11. https://doi.org/10.1038/s41598-024-84695-4